

Saroj Khadka

Graduate Student Researcher

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Summary

Biomedical Science PhD candidate with strong research experience in bacterial virulence mechanisms, host-pathogen interactions, and microbial gene regulation. Skilled in molecular biology and microbiological techniques, mentoring and science communication. Motivated to build an independent career in biomedical research.

Education

- PhD University of Pittsburgh**, PhD in Microbiology and Immunology – Pennsylvania, PA, 2024 – 2026
USA
- Thesis: Mechanisms and Regulation of *Klebsiella pneumoniae* Capsular Polysaccharide Chain Length and Attachment.
 - Advisor: Dr. Laura A. Mike
- MSc Tribhuvan University**, MSc in Medical Microbiology – Kathmandu, Nepal 2017 – 2019
- Thesis: Susceptibility to Fluoroquinolones and *gyrA* ser83 Mutation in *Salmonella enterica* serovar Typhi at a Referral Hospital of Kathmandu Valley.
 - Advisor: Dr. Megharaj Banjara

Experience

- University of Pittsburgh**, Graduate Student Researcher – Pennsylvania, PA, USA 2024 – 2026
- Investigate molecular mechanisms regulating *Klebsiella pneumoniae* capsular polysaccharide attachment and chain length.
 - Develop and optimize experimental techniques for bacterial exopolysaccharide research.
 - Present research findings at departmental, regional, and national scientific conferences.
 - Mentor undergraduate researchers in laboratory techniques, experimental workflows, and project design.
- B-Sure Path Lab & Diagnostic Centre**, Molecular Biologist – Biratnagar, Nepal 2020 – 2021
- Performed routine qRT-PCR testing for COVID-19 diagnosis from clinical specimens.
 - Trained newly recruited laboratory technicians in molecular diagnostic techniques and workflows.
 - Managed patient records, diagnostic reporting, and laboratory quality control in compliance with National Public Health Laboratory (Nepal) guidelines.
 - Coordinated diagnostic reporting with the Ministry of Health COVID-19 Crisis Management Center.
- Newton Higher Secondary School**, Secondary School Teacher – Kathmandu, Nepal 2018 – 2019
- Taught Science and Mathematics to Grade 8 students.
 - Designed and administered tests, and prepared test score reports to track student progress.
 - Facilitated parent-teacher meetings to discuss student development and address concerns.
 - Proctored final examinations and National Education Board exams for Grade 8 and Grade 10 students.
- Kathmandu Model Hospital**, Intern Medical Microbiologist – Kathmandu, Nepal 2018 – 2019
- Conducted routine clinical microbiological tests for pathogen isolation and identification in clinical specimens.
 - Maintained culture collection and surveillance records of multidrug-resistant bacterial isolates for national reporting.
 - Performed monthly and quarterly quality assurance assays as required by National Public Health Laboratory.
 - Performed hospital environmental pathogen monitoring.

Awards

- MMI Research Fellowship Award** 2024
- Awarded by the Department of Medical Microbiology and Immunology, University of Toledo to graduate students who have received a pre-doctoral fellowship before thesis defense.
- MMI

MMI Research Scholar Award

2024

Awarded by the Department of Medical Microbiology and Immunology, University of Toledo to graduate students with a first-author research manuscript accepted or published before thesis defense.

MMI

AHA Predoctoral Fellowship

2023

Two-year fellowship awarded by the American Heart Association (AHA) in the form of stipend support to PhD students pursuing research aimed at advancing human health and well-being.

MMI

Graduate Research Award

2024

Awarded by the Department of Medical Microbiology and Immunology, University of Toledo to graduate students with a first-author research manuscript accepted or published before thesis defense.

MMI

Publications

Zur Elektrodynamik bewegter Körper

It concerned an interpretation of the Michelson–Morley experiment and the properties of light and time. Special relativity incorporates the principle that the speed of light is the same for all inertial observers regardless of the state of motion of the source.

Albert Einstein

en.wikisource.org/wiki/Translation:On_the_Electrodynamics_of_Moving_Bodies**Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen****Gesichtspunkt**

In the second paper, he applied the quantum theory to light to explain the photoelectric effect. In particular, he used the idea of light quanta (photons) to explain experimental results, but stressed the importance of the experimental results. The importance of his work on the photoelectric effect earned him the Nobel Prize in Physics in 1921.

Albert Einstein

[de.wikisource.org/](https://de.wikisource.org/wiki/%C3%9Cber_einen_die_Erzeugung_und_Verwandlung_des_Lichtes_betreffenden_heuristischen_Gesichtspunkt)[wiki/%C3%9Cber_einen_die_Erzeugung_und_Verwandlung_des_Lichtes_betreffenden_heuristischen_Gesichtspunkt](https://de.wikisource.org/wiki/%C3%9Cber_einen_die_Erzeugung_und_Verwandlung_des_Lichtes_betreffenden_heuristischen_Gesichtspunkt)**Die Grundlage der allgemeinen Relativitätstheorie**

The publication of the theory of general relativity made him internationally famous. He was professor of physics at the universities of Zurich (1909–1911) and Prague (1911–1912), before he returned to ETH Zurich (1912–1914).

Albert Einstein

de.wikisource.org/wiki/Die_Grundlage_der_allgemeinen_Relativit%C3%A4tstheorie

Skills

Physics

Languages

Nepali

Native speaker

English

Fluent

Certificates

Machine Learning

Jan 2018

Projects

Systems controlling *Klebsiella pneumoniae* capsular polysaccharide chain length and attachment

2022 – present

Currently investigating the mechanisms and regulatory pathways governing *Klebsiella pneumoniae* capsule attachment and chain length using various molecular biology and biochemical analytical tools. The ongoing projects aim to unravel the bacterial factors and signaling pathways involved in *K. pneumoniae* capsular polysaccharide attachment and chain length regulation and their impact on bacterial pathogenesis.

- Advisor: Dr. Laura A. Mike

Susceptibility to Fluoroquinolones and *gyrA* ser83 mutation in *Salmonella enterica* Serovar Typhi at a Referral Hospital of Kathmandu Valley

2018 – 2019

Conducted a hospital-based study on the prevalence of *Salmonella* Typhi in febrile patients visiting the Kathmandu Model Hospital and evaluated resistance to fluoroquinolones associated with *gyrA* ser83 mutation in *Salmonella* Typhi and Paratyphi isolated from the patients.

- Advisor: Dr. Megharaj Banjara

References

Dr. Laura A. Mike

Dr. Megharaj Banjara